How Do Large Language Models Count in French? Mapping Robust Transduction and Probe-Sensitive Irregularity

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Abstract

French number words are compositionally systematic but atypical, especially in the 70–99 range (e.g., *soixante-dix*, *quatre-vingt-dix*). We test whether current large language models show measurable weakness on these irregular forms. We introduce FRCOUNTMAP, a reproducible black-box probe pipeline that evaluates numeral transduction and pattern-level classification with deterministic decoding. Our synthetic suite covers every integer from 0 to 199, and we complement it with MGSM-FR for broader mathematical reasoning.

Across 1,935 analyzed outputs, both models achieve near-ceiling performance on direct transduction. For GPT-4.1, DIGIT→WORD reaches 98.6%–100% across all bands and WORD→DIGIT is 100% in every complete condition. GPT-4.1-MINI remains similarly strong (96.7%–100% on DIGIT→WORD; 100% on WORD→DIGIT). Two-proportion tests show no significant irregular-band penalty for transduction after FDR correction. The strongest band effect appears in PATTERN-CLASSIFICATION, where irregular items are easier under our label schema (e.g., risk difference $+0.435$, FDR-corrected $p < 0.001$ for GPT-4.1 direct). On MGSM-FR, reasoning prompts improve GPT-4.1 from 48.8% to 57.5%.

These results show that modern API LLMs can reliably map French numerals under low-temperature prompting, while conclusions about "difficulty" depend strongly on probe design.

1 Introduction

French counting is a clean stress test for multilingual numeracy: it is rule-based, but its lexical surface is atypical in the 70–99 range. Errors in numeric language matter in tutoring, translation, and safety-critical communication. Our main question is simple: do strong LLMs degrade on French irregular forms, or can they map these forms consistently?

Prior numeracy studies establish that tokenization, representation choice, and synthetic pretraining all affect number handling [Wallace et al., 2019, Geva et al., 2020, Jin et al., 2021]. Multilingual reasoning benchmarks such as MGSM report aggregate French performance [Shi et al., 2022], and interpretability work identifies arithmetic-sensitive components [Yu and Ananiadou, 2024, Chang et al., 2024]. However, existing evaluations do not isolate French-specific numeral morphology with full coverage and band-level significance tests.

We address this gap with FRCOUNTMAP, an inference-only probe suite for French numerals. We evaluate deterministic API outputs on DIGIT→WORD, WORD→DIGIT, and PATTERN-CLASSIFICATION tasks over a synthetic 0–199 dataset, then connect those results to MGSM-FR performance. Our quantitative preview is direct: transduction is near-perfect and does not show a significant irregular-band penalty, while PATTERN-CLASSIFICATION exhibits large and significant irregular-band advantages under our label design.

In summary, our main contributions are:

- We introduce a reproducible French numeral probe pipeline with complete 0–199 coverage and transparent normalization/scoring.
- We conduct controlled model-by-condition comparisons showing ceiling-level WORD→DIGIT and near-ceiling DIGIT→WORD performance without a robust irregular-range drop.
- We provide band-wise risk-difference tests with FDR correction and a pooled logistic model that quantifies where band effects do and do not appear.
- We show that prompt reasoning affects broader French math reasoning (MGSM-FR) more than numeral transduction itself.

Paper organization. Section 2 discusses prior numeracy and multilingual reasoning research. Section 3 details data, probes, and metrics. Section 4 presents quantitative findings and statistical tests. Section 5 covers interpretation, limitations, and implications, and section 6 concludes.

2 Related Work

Numeracy in language representations. Early probing work shows that standard embeddings often encode numbers weakly outside seen ranges, with large extrapolation gaps [Wallace et al., 2019]. Follow-up approaches inject numeric skills with synthetic data or numeral-aware objectives, yielding stronger numerical reasoning in downstream tasks [Geva et al., 2020, Jin et al., 2021]. Our work differs in scope: we do not train or adapt models; we isolate French numeral behavior through controlled black-box probes.

Multilingual mathematical reasoning. MGSM established that chain-of-thought prompting can improve multilingual grade-school math, including French [Shi et al., 2022]. Subsequent multilingual math instruction work reports additional gains from cross-lingual training data [Chen et al., 2024]. These studies evaluate broad reasoning, while we focus on a narrower linguistic bottleneck: mapping French numeral strings with explicit banded analysis.

Tokenization and language-specific structure. French tokenization design materially changes downstream behavior in domain models [Labrak et al., 2024]. This aligns with our motivation to inspect French numeral morphology directly rather than rely only on aggregate benchmark scores.

Mechanistic views of arithmetic. Recent analyses identify arithmetic-sensitive heads/neurons and argue that structural priors support generalization [Yu and Ananiadou, 2024, Chang et al., 2024]. We complement this line by delivering a task suite that can be used as a front-end diagnostic before token-level or activation-level mechanistic tracing.

3 Methodology

Overview. We run deterministic, inference-only API probes with temperature $= 0$ to measure functional representation quality without model retraining. We evaluate two models (GPT-4.1, GPT-4.1-MINI), two prompt conditions (direct, reasoned), and four tasks: DIGIT→WORD, WORD→DIGIT, PATTERN-CLASSIFICATION, and MGSM-FR.

Datasets. Our core dataset is a synthetic French numeral set covering all integers from 0 to 199, generated by canonical rules in `src/french_numerals.py`. This yields 200 unique examples with no duplicates. We partition by number band: 0–69, 70–99, and 100–199. For broader reasoning, we evaluate a French MGSM subset (`datasets/mgsm/mgsm_fr.tsv`) with 250 rows; missing question rate is 0%, missing answer rate is 1.6%, and no duplicate questions were detected.

Scoring and normalization. For DIGIT→WORD, we score exact and normalized matches after lowercasing and hyphen/spacing normalization. For WORD→DIGIT, we parse predicted integers using regex and score exact integer match. PATTERN-CLASSIFICATION is scored by exact class label match. We treat 70–99 as the irregular band and aggregate 0–69 plus 100–199 as regular for significance tests.

Statistical analysis. We compute accuracy by task, band, model, and condition. For each complete run, we test irregular-vs-regular differences with two-proportion z-tests, report risk differences and

Model	Condition	Task	0–69	70–99	100–199
GPT-4.1	Direct	DIGIT→WORD	0.9857	**1.0000**	**1.0000**
GPT-4.1	Reasoned	DIGIT→WORD	0.9857	**1.0000**	**1.0000**
GPT-4.1-MINI	Direct	DIGIT→WORD	0.9714	0.9667	**1.0000**
GPT-4.1	Direct	WORD→DIGIT	**1.0000**	**1.0000**	**1.0000**
GPT-4.1	Reasoned	WORD→DIGIT	**1.0000**	**1.0000**	**1.0000**
GPT-4.1-MINI	Direct	WORD→DIGIT	**1.0000**	**1.0000**	**1.0000**

Table 1: Band-wise transduction accuracy. Best values within each task block are in **bold**. GPT-4.1-MINI reasoned was incomplete and is excluded.

Model	Task	Risk diff.	p	FDR sig.
GPT-4.1 (direct)	DIGIT→WORD	+0.0059	0.6737	No
GPT-4.1-MINI (direct)	DIGIT→WORD	-0.0216	0.3702	No
GPT-4.1 (direct)	PATTERN-CLASSIFICATION	+0.4353	5.29e-06	Yes
GPT-4.1 (reasoned)	PATTERN-CLASSIFICATION	+0.4235	8.36e-06	Yes
GPT-4.1-MINI (direct)	PATTERN-CLASSIFICATION	+0.4805	1.58e-06	Yes

Table 2: Irregular-vs-regular band tests. Positive risk difference indicates higher accuracy on 70–99.

confidence intervals, and apply Benjamini–Hochberg FDR correction at $q = 0.05$. We also fit a pooled logistic model:

$$\texttt{correct} \sim \texttt{irregular} + C(\texttt{model}) + C(\texttt{condition}) + C(\texttt{task}).$$

Implementation details. We use Python 3.12.8 with `openai` 2.26.0, `pandas` 3.0.1, `numpy` 2.4.2, `scipy` 1.17.1, `statsmodels` 0.14.6, `matplotlib` 3.10.8, and `seaborn` 0.13.2. Generation uses `temperature=0`, `max_output_tokens=180`, and `seed=42`. Raw outputs are logged in `results/raw/raw_outputs_20260308_214204.jsonl`.

Baselines. Our principal baseline is the direct prompt condition; the reasoned condition is an ablation on output style. We do not include training-based baselines because the study targets black-box API behavior under controlled prompting.

4 Results

Aggregate performance. We analyze 1,935 rows with overall accuracy 0.835 and zero API error rows. The lower overall score is driven by harder tasks (PATTERN-CLASSIFICATION and MGSM-FR), not by numeral transduction.

Core transduction findings. Table 1 shows near-ceiling results on DIGIT→WORD and ceiling-level results on WORD→DIGIT. GPT-4.1 reaches 98.6%–100% on DIGIT→WORD and 100% in all WORD→DIGIT bands and conditions. GPT-4.1-MINI direct remains high (96.7%–100% on DIGIT→WORD; 100% on WORD→DIGIT). These results do not support the predicted irregular-band degradation for transduction.

Band-effect statistics. Table 2 summarizes irregular-vs-regular tests. For DIGIT→WORD, neither GPT-4.1 direct ($+0.0059$, $p = 0.6737$) nor GPT-4.1-MINI direct (-0.0216, $p = 0.3702$) is significant after FDR correction. In contrast, PATTERN-CLASSIFICATION shows large positive irregular-band effects in every complete run (all FDR-corrected $p < 0.001$).

Pooled logistic regression. In the pooled core-probe model ($n = 1775$), the irregular-band coefficient is positive and significant ($p \approx 1.19 \times 10^{-9}$), while reasoned-vs-direct is not significant ($p \approx 0.83$). This confirms that band effects exist in the combined probe space, but they are task-dependent rather than a uniform transduction failure.

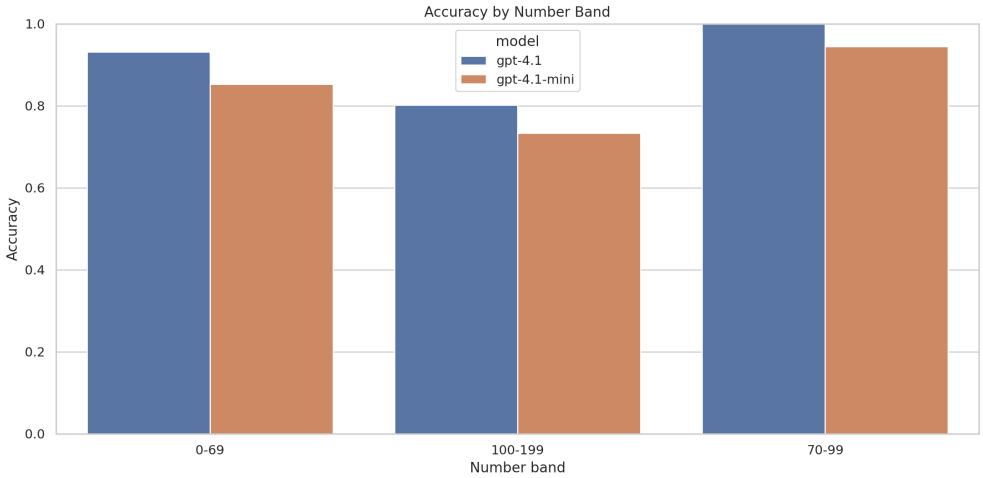

Figure 1: Accuracy by number band, model, and task. Transduction tasks stay near ceiling, while PATTERN-CLASSIFICATION shows larger band separation.

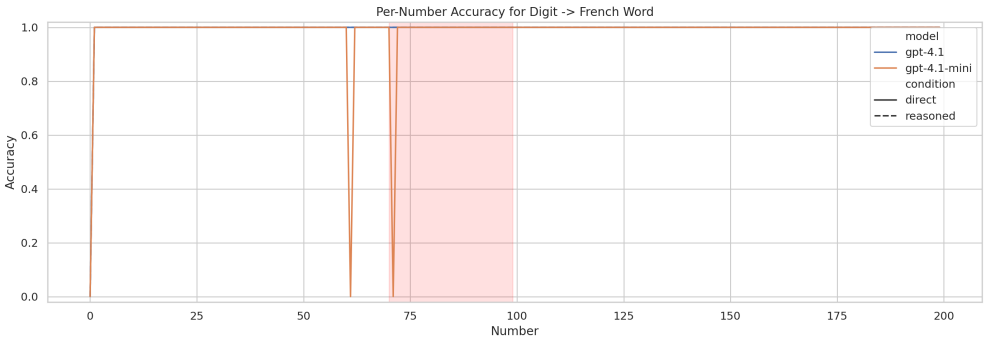

Figure 2: Per-number DIGIT→WORD accuracy. Errors are sparse and not concentrated in the 70–99 range.

Prompt ablation on MGSM-FR. Reasoned prompting improves GPT-4.1 from 48.8% to 57.5% on MGSM-FR, an absolute gain of 8.75 points. This is substantially larger than condition effects in core transduction probes.

Error analysis. Representative failures are mostly orthographic variants rather than numeric misunderstandings. For example, for 71 the model output *soixante-et-onze* instead of canonical *soixante et onze*; normalization resolves many such cases.

5 Discussion

What do these results imply? The main empirical point is separation between capabilities. French numeral transduction appears robust for current strong API models under deterministic prompts, but probe framing can create large apparent band effects. Our PATTERN-CLASSIFICATION results should therefore be interpreted as evidence about label-space interactions, not as direct proof of weaker French numeral representation.

Why does PATTERN-CLASSIFICATION favor irregular items? Our label schema likely makes irregular constructions more discriminative, which increases separability. This is useful diagnostically, but it can overstate "difficulty" claims if used in isolation. The contrast with near-perfect DIGIT→WORD and WORD→DIGIT performance supports this interpretation.

Limitations. One model-condition block (GPT-4.1-MINI reasoned) is incomplete because of runtime interruption, so the factorial design is not fully balanced. We also do not include token-level log-probabilities or activation-level analysis, which limits mechanistic conclusions. Finally, our canonicalization remains conservative and may still mark legitimate French orthographic variants as errors in strict mode.

Broader implications. For deployment, the positive result is practical reliability in low-temperature numeral mapping. For evaluation design, the caution is equally important: benchmark conclusions about language-specific structure can be probe-sensitive. Future multilingual numeracy benchmarks should separate formatting sensitivity from true semantic errors and report both strict and normalized metrics.

6 Conclusion

We presented FRCOUNTMAP, a reproducible probe framework for French numeral representation. Across 1,935 analyzed outputs, DIGIT→WORD and WORD→DIGIT are near-ceiling to ceiling, with no significant irregular-band penalty for completed transduction conditions. Significant band effects emerge mainly in PATTERN-CLASSIFICATION, and reasoning prompts matter more for MGSM-FR than for core numeral transduction.

The key takeaway is that modern LLMs can reliably map French numeral forms in deterministic settings, but claims about representation difficulty depend strongly on probe design.

Next steps are straightforward: complete the missing GPT-4.1-MINI reasoned block, expand normalization to explicitly include orthographic reform variants, add mechanistic probes on open models, and test out-of-distribution variants such as Belgian/Swiss forms (e.g., *septante*, *nonante*).

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